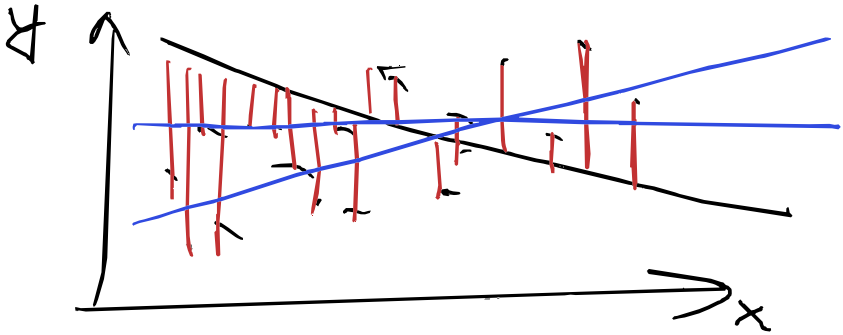


OZNAČENI PODACI  $(x^{(i)}, y^{(i)})$



$$h(\theta) = \theta_0 + \theta_1 x$$

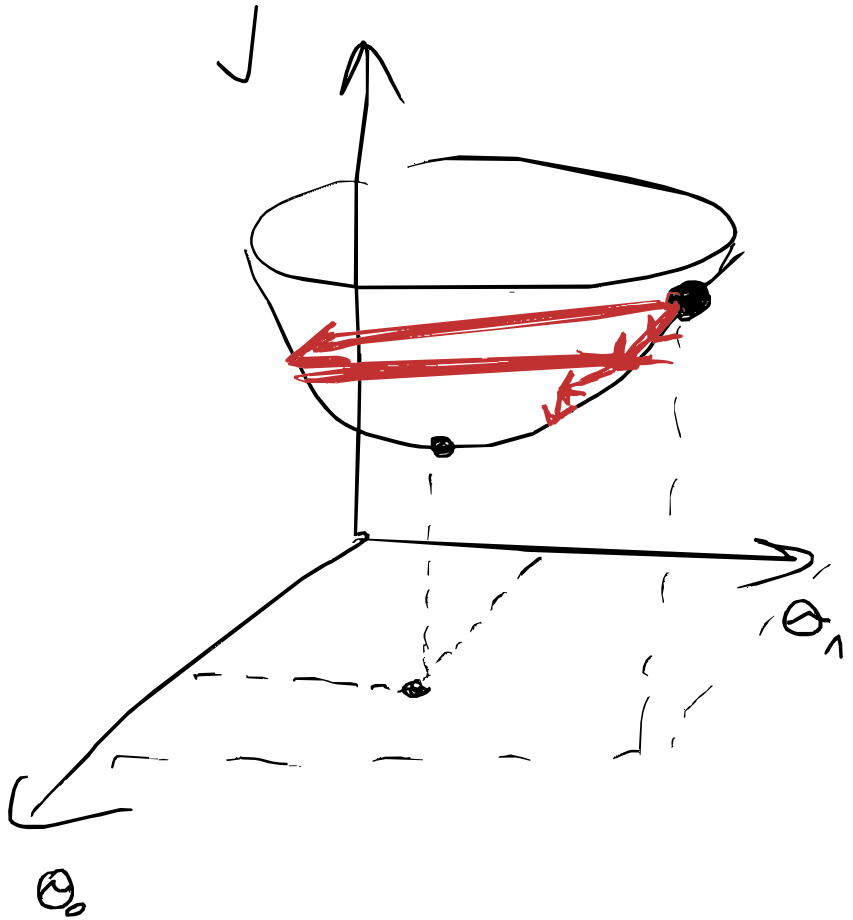
$J \Rightarrow$  kvadratična FUNKCIJA  
(MSE)

CILJ  $\Rightarrow$  MINIMIZIRATI GREŠKU

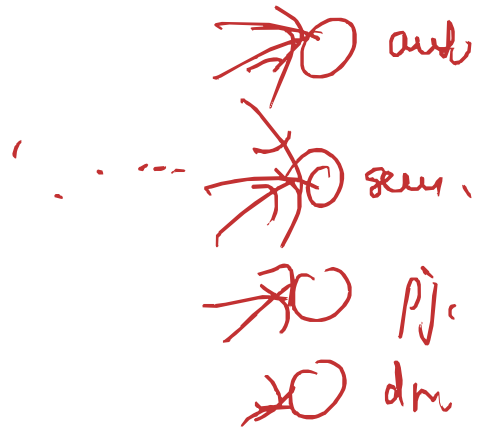
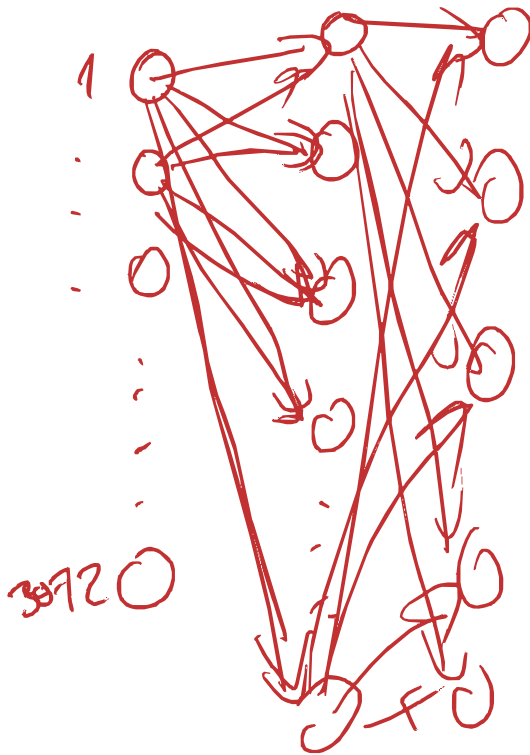
LOSS = GREŠKA

$$\theta(t+1) = \theta(t) - \alpha \cdot \frac{\partial J}{\partial \theta}$$

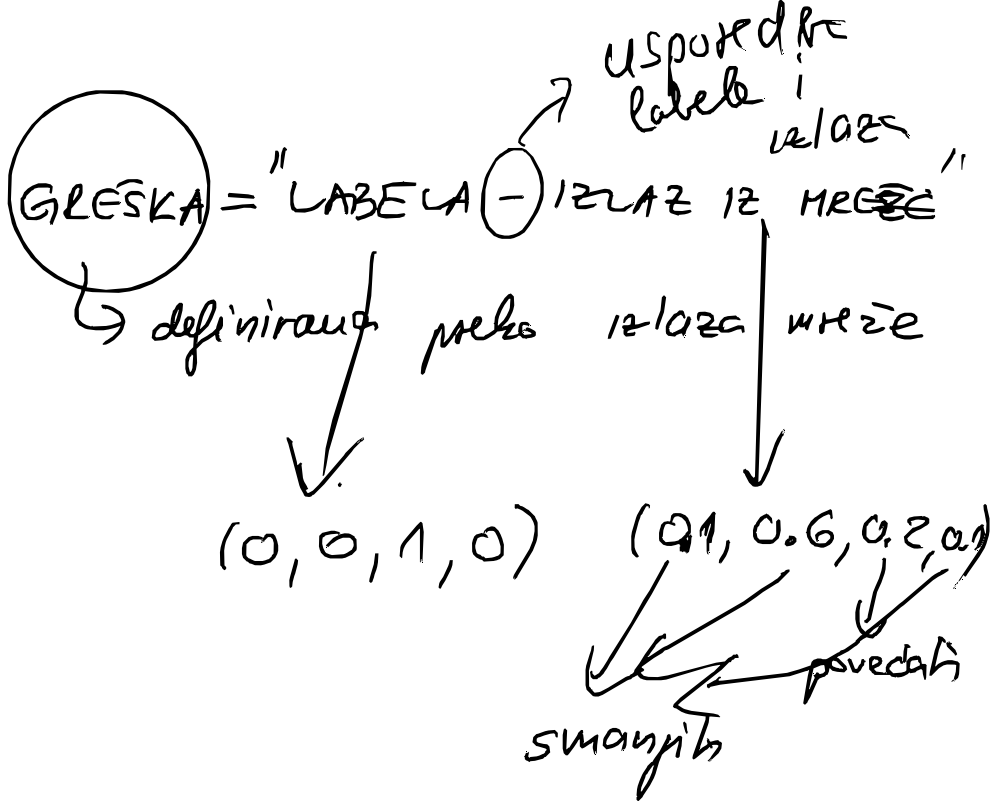
LEARNING RATE, BRZINA UČENJA,  
STOPA UČENJA



Learning rate  $\Rightarrow$  KOLIKO VELIKE  
KORAKE PRILIKOM SPUSTANJA DA  
RAZIM?



PRAZNA  
STRANICA



ZNAM DA MORAM POVEČATI 3. IZLAZ  
IZ MREŽE DA BI SE GREŠKA  
SMANJILA

BATCH

MINI-BATCH

EROTHA

ITERACIJA

---

Imate na raspolaganju 1000  
označenih podatka.

---

Podijelite skup na 3 podskup

- TRAINING 800

- VALIDACIJA 100

- TEST 100

Tr val. Test

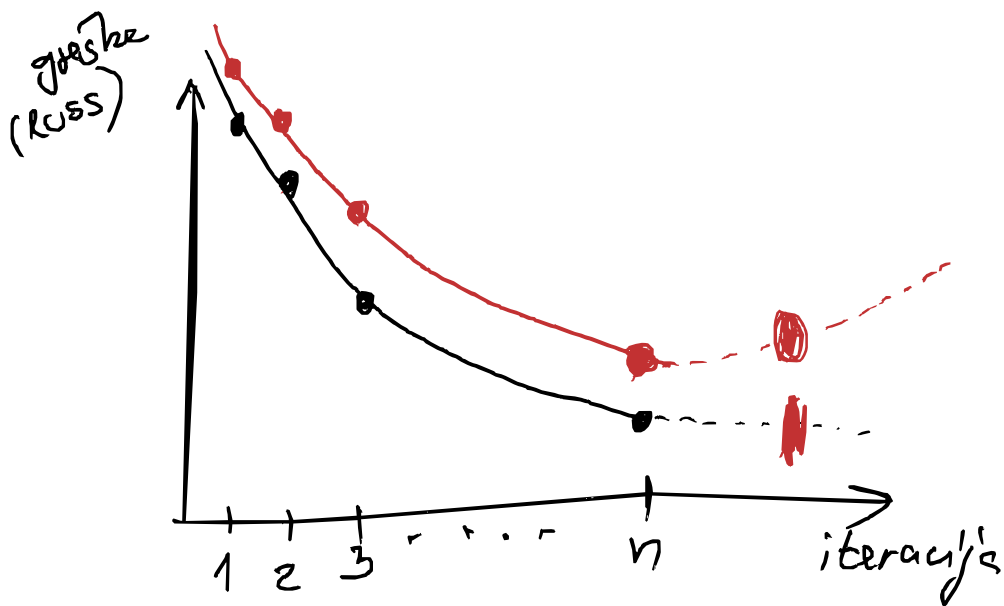
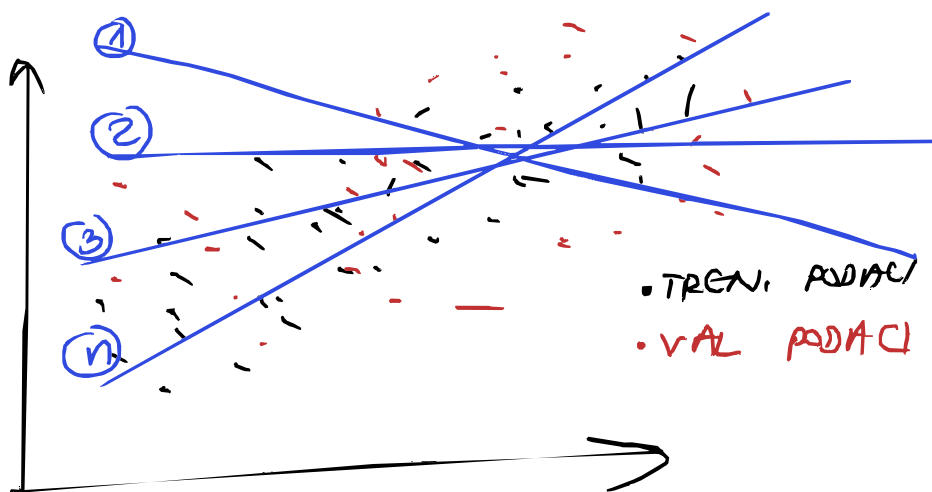
80 : 10 : 10

70 : 20 : 10

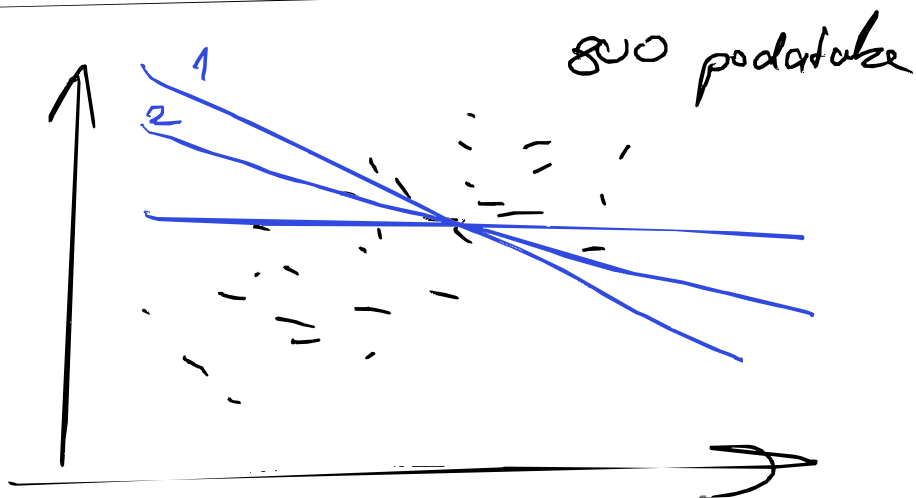
70 : 15 : 15

---

npr. imamy 800 trening poddanych  
i 100 validacyjnych



KADA ZA JEDNU PROMJENU  
PARAMETARA MODELA/MREŽE  
GREŠKU RAČUNAM KORISTEĆI  
SVE TRJNING PODATKE, TO SE  
ZOVE **BATCH METODA.**



npr. uzmi 80 od tih 800  
i izračunaj grešku  
- uzmi sljedećih 80 podataka i  
izračunaj grešku za 2. put



TREBA MI 10 ITERACIJA  
PROMJENA PARAMETARA MODELA/  
MREŽE DA BIM JEDNOM PROŠAO  
KROZ SVE TRAINING PODATKE?

↳ TO SE ZOVE **MINI-BATCH  
METODA**

1 PROLAZAK KROZ SVE TRAINING  
PODATKE NAZIVA SE EPOHA  
TRAINING

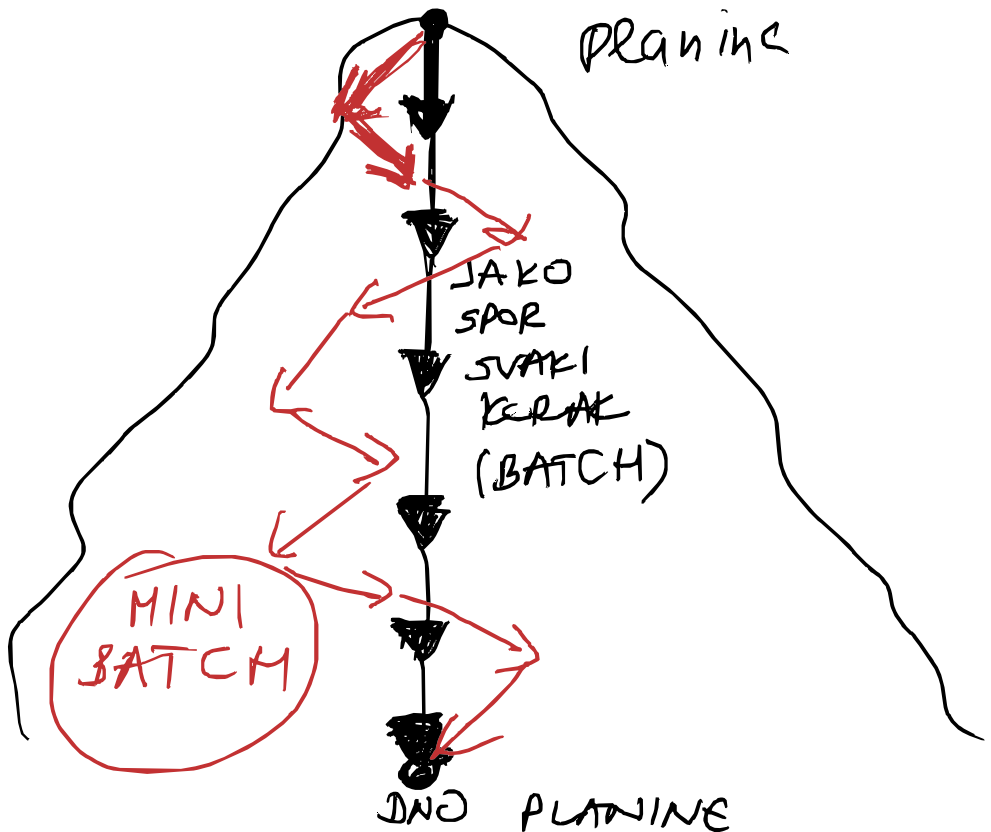
Zadatak

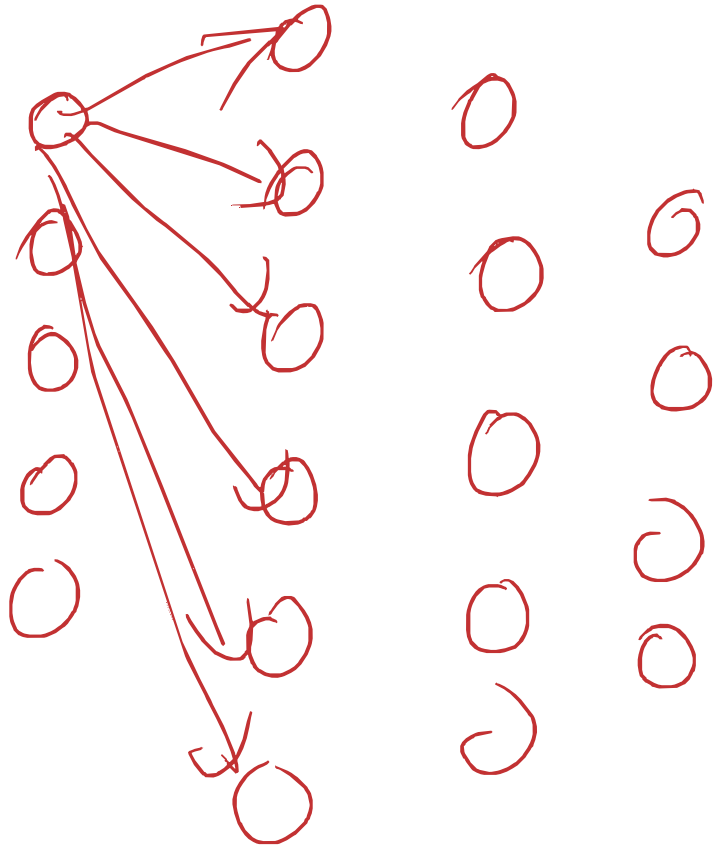
Provedi training na 800 podataka  
u 20 epoha, uz mini-batch  
vrijednost 8. Koliko ti iteracija's  
traininga treba?

Marko 2000 ✓

PAZITI  $\Rightarrow$  MINI-BATCH JE  
PARAMETAR KOJI SE  
ZOVE "BATCH SIZE"

---



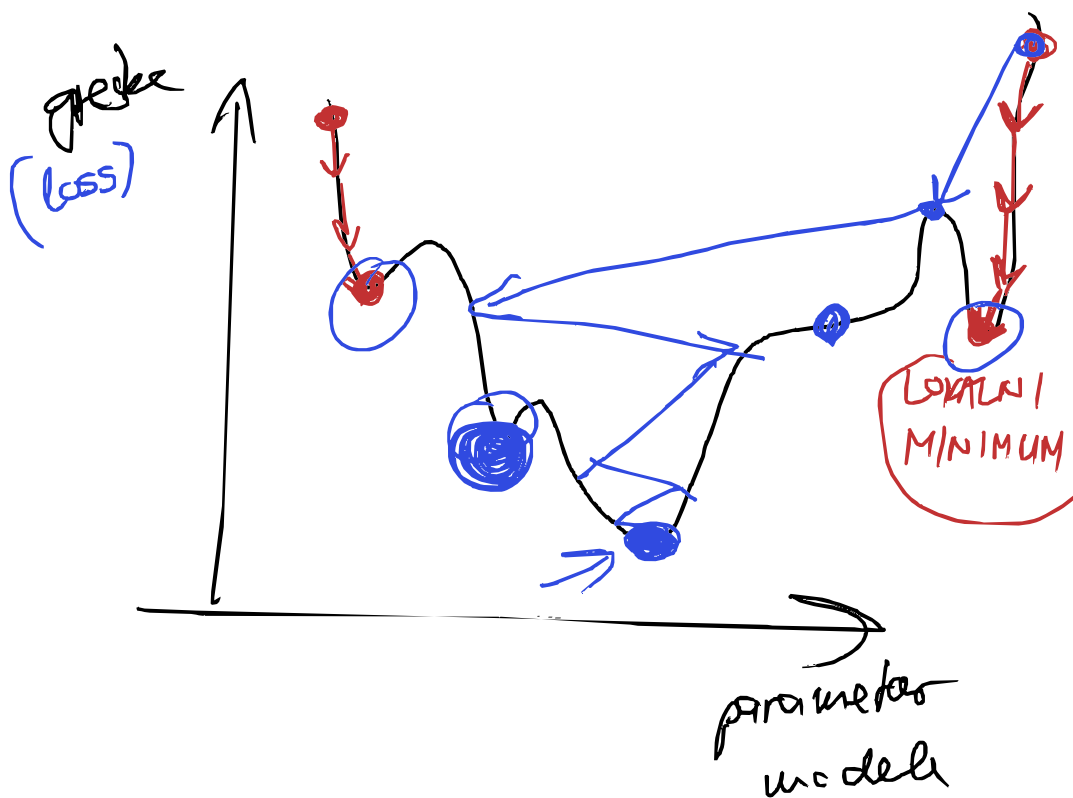


SUAKA STRELIKA JE JEDAN PARAMETER

U PRAKSI :

BATCH-SIZE 16, 32, 64,  
128, 256, 512,  
...

SE POKAZAO NAJBOLJI



LEARNING RATE

$\alpha$

0,001 - 0,000001

$10^{-3}$  -  $10^{-6}$

U PRAKSI

KASNIJE  $\Rightarrow$  SMANJIVANJE  $\alpha$

KROZ EPHE TRENINGA

# DROPOUT

↳ TEHNIKA ZA SPREJEDJIVANJE  
OVERFITTINGA

